

A LOCKED STRIP-NORM REDUCTION OF OPERATORS ON $L_P(L_1)$ TO BI-PARAMETER HAAR MULTIPLIERS

FULL PROOF DRAFT V32

ABSTRACT. Let $1 < p < \infty$ and $X = L_p([0, 1]; L_1[0, 1])$. This draft gives a detailed proof that every bounded operator on the doubly cancellative product-Haar part X_{00} has a locked faithful-Haar compression which is arbitrarily close to a bounded bi-parameter Haar multiplier. The faithful systems are locked coordinatewise: one outer tree and one inner tree are constructed once and for all, so the proof avoids the false refreshed-window route along L_1 strips. The off-product-diagonal part is estimated by a strip-norm decomposition: same-outer/different-inner strips are outer L_p multipliers, different-outer/same-inner strips are inner L_1 multipliers controlled by Semenov–Uksusov branch variation, and fully mixed terms are made absolutely summable. The local L_1 trace theorem used in the inner strip argument is derived in the text from the Semenov–Uksusov multiplier characterization. Combining the arbitrary-to-multiplier reduction with the multiplier theorem of Lechner–Motakis–Muller–Schlumprecht gives scalar compressions; the standard Rademacher-doubling reduction from $L_p(L_1)$ to X_{00} then gives primarity.

1. SET-UP AND REDUCTION TO THE PRODUCT-CANCELLATIVE PART

Throughout the paper $1 < p < \infty$ is fixed and

$$X = L_p([0, 1]; L_1[0, 1]).$$

We work over the real scalar field. The complex case is obtained by inserting conjugates in the coefficient pairings and by using the complex forms of the cited multiplier theorems.

Let $\mathcal{D}$ be the dyadic intervals in $[0, 1)$. For $I \in \mathcal{D}$, h_I denotes the usual $\{-1, 1\}$ -valued Haar function supported on I . Put

$$X_{00} = \overline{\text{span}}\{h_I \otimes h_J : I, J \in \mathcal{D}\} \subset X.$$

This is the doubly cancellative product-Haar part of X . We use the normalized product vectors

$$e_I = |I|^{-1/p} h_I, \quad e_I^* = |I|^{-1/p'} h_I,$$

for the outer coordinate and

$$f_J = |J|^{-1} h_J, \quad f_J^* = h_J,$$

for the inner coordinate. Thus $\|e_I\|_p = \|e_I^*\|_{p'} = 1$, $\|f_J\|_1 = \|f_J^*\|_\infty = 1$, and

$$u_{I,J} = e_I \otimes f_J, \quad u_{I,J}^* = e_I^* \otimes f_J^*.$$

Throughout, functionals on the zero-mean spaces L_q^0 are represented by concrete functions modulo constants. Whenever such a representative is used in an estimate, we choose one with norm within an arbitrarily small factor of the quotient norm; this harmless convention is suppressed below.

Let $\mathbb{E}_1$ and $\mathbb{E}_2$ denote integration in the first and second variables. Then

$$P_{00} = (\text{Id} - \mathbb{E}_1)(\text{Id} - \mathbb{E}_2)$$

is a bounded projection from X onto X_{00} with $\|P_{00}\| \leq 4$.

Lemma 1.1 (A one-complemented cancellative copy). *The space X_{00} contains an isometric copy of X which is the range of a norm-one projection on X .*

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Proof. Let

$$r(t) = \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t), \quad \sigma(t) = 2t \pmod{1}.$$

Define

$$(Uf)(s, t) = r(s)r(t)f(\sigma(s), \sigma(t)).$$

Since σ preserves Lebesgue measure and $|r| = 1$, U is an isometry on X . Moreover, $\mathbb{E}_1 Uf = \mathbb{E}_2 Uf = 0$, so $U(X) \subset X_{00}$.

Define

$$(Vg)(u, v) = \frac{1}{4} \sum_{\alpha, \beta \in \{0,1\}} (-1)^{\alpha+\beta} g\left(\frac{u+\alpha}{2}, \frac{v+\beta}{2}\right).$$

Then $VU = \text{Id}_X$. If $a(s) = \|g(s, \cdot)\|_{L_1}$, then

$$\|Vg(u, \cdot)\|_{L_1} \leq \frac{1}{2} (a(u/2) + a((u+1)/2)).$$

Convexity and a change of variables give $\|Vg\|_{L_p(L_1)} \leq \|g\|_{L_p(L_1)}$. Hence UV is a norm-one projection of X onto $U(X) \subset X_{00}$. $\square$

Proposition 1.2 (Scalar compression on X_{00} implies primarity). *Assume that for every $S \in \mathcal{L}(X_{00})$ and every $\eta > 0$ there are operators $A_0, B_0 \in \mathcal{L}(X_{00})$ and a scalar c such that*

$$A_0 B_0 = \text{Id}_{X_{00}}, \quad \|A_0 S B_0 - c \text{Id}_{X_{00}}\| < \eta.$$

Then $X = L_p(L_1)$ is primary.

Proof. Let $T \in \mathcal{L}(X)$ and set $S = P_{00}T|_{X_{00}} \in \mathcal{L}(X_{00})$. By hypothesis, for every $\eta > 0$ there are A_0, B_0 and c with

$$A_0 B_0 = \text{Id}_{X_{00}}, \quad \|A_0 P_{00} T B_0 - c \text{Id}_{X_{00}}\| < \eta.$$

Thus scalar compressions on X_{00} give scalar compressions of arbitrary operators on X , with $A = A_0 P_{00}$ and $B = B_0$.

Apply this to a projection P on X , and choose $\eta < 1/4$. Either $|c| \geq 1/2$ or $|1-c| \geq 1/2$. In the first case APB is invertible on X_{00} and

$$\text{Id}_{X_{00}} = (APB)^{-1} APB$$

factors through P . In the second case

$$A(\text{Id}_X - P)B = \text{Id}_{X_{00}} - APB$$

is invertible and

$$\text{Id}_{X_{00}} = (A(\text{Id}_X - P)B)^{-1} A(\text{Id}_X - P)B$$

factors through $\text{Id}_X - P$. Let $U : X \rightarrow X_{00}$ and $V : X \rightarrow X$ be the maps from Lemma 1.1, with $VU = \text{Id}_X$ and with V restricted to X_{00} when it is composed on the left. Thus in the first case, for example,

$$\text{Id}_X = V(APB)^{-1} APBU,$$

so Id_X factors through P ; the second case is identical with P replaced by $\text{Id}_X - P$. Finally, X is isomorphic to the ℓ_p -sum of countably many copies of itself by splitting the outer variable into countably many dyadic pieces. Pełczyński's decomposition method therefore turns this primary factorization property into primarity. $\square$

2. EXTERNAL INPUTS AND FAITHFUL EXTERIOR MAPS

We isolate the standard outside ingredients used later.

Theorem 2.1 (Semenov–Uksusov branch variation on L_1). *Let D be a scalar Haar multiplier on L_1^0 , $Dh_I = a_I h_I$. Then D is bounded on L_1^0 if and only if*

$$\text{BV}_{\text{br}}(a) = \sup_{I_0 \supset I_1 \supset \dots} \left(|a_{I_0}| + \sum_{k \geq 0} |a_{I_{k+1}} - a_{I_k}| + \limsup_k |a_{I_k}| \right) < \infty,$$

where the supremum is over dyadic branches. Moreover

$$\|D : L_1^0 \rightarrow L_1^0\| \approx \text{BV}_{\text{br}}(a)$$

with absolute constants.

Proposition 2.2 (Local L_1 trace theorem). *Let $R \in \mathcal{L}(L_1^0)$ and let B be a finite equal-dyadic atom. If $\mathcal{D}_n(B)$ denotes the equal-measure dyadic descendants of B at relative depth n , then*

$$\beta_n^B(R) = \frac{1}{\#\mathcal{D}_n(B)} \sum_{J \in \mathcal{D}_n(B)} \langle f_J^*, R f_J \rangle$$

converges as $n \rightarrow \infty$. We denote the limit by $\lambda_B(R)$. There is a function $\varphi_R \in L_\infty[0, 1]$ with $\|\varphi_R\|_\infty \leq C \|R\|$ such that

$$\lambda_B(R) = \frac{1}{|B|} \int_B \varphi_R$$

for every dyadic interval B , and hence for every finite equal-dyadic atom B by additivity.

Proof. We first treat the case in which B is a dyadic interval. Rescale B to $[0, 1]$, and let P_N^B be the martingale-difference projection onto the local Haar levels $0, 1, \dots, N$ below B . Explicitly, if $\mathcal{A}_{N+1}(B)$ is the partition of B into relative-depth $N + 1$ dyadic subintervals and E_{N+1}^B is conditional expectation on this partition, applied to $\mathbf{1}_B x$, then

$$P_N^B x = E_{N+1}^B(\mathbf{1}_B x) - \frac{1}{|B|} \left(\int_B x \right) \mathbf{1}_B.$$

Hence $\|P_N^B\|_{L_1 \rightarrow L_1} \leq 2$, uniformly in B and N , and P_N^B maps into L_1^0 .

Let

$$V_N(B) = \text{span}\{f_J : J \in \mathcal{D}_k(B), 0 \leq k \leq N\}.$$

The projection P_N^B is the biorthogonal Haar projection onto $V_N(B)$,

$$P_N^B x = \sum_{k=0}^N \sum_{J \in \mathcal{D}_k(B)} \langle f_J^*, x \rangle f_J,$$

and has uniformly bounded norm on L_1^0 . Let $G_N(B)$ be the finite group of measure-preserving dyadic automorphisms of the finite dyadic tree below B whose terminal atoms have relative depth $N + 1$, preserving relative levels. The extra terminal level is included so that a Haar function whose support has relative level N can still be sign-flipped by swapping the two children of its support. Each finite-tree automorphism is extended below the terminal atoms by the canonical level-preserving dyadic affine maps, so it induces an isometry U_γ on all of L_1^0 . For every local Haar node J of level at most N , these isometries satisfy

$$U_\gamma f_J = \varepsilon_\gamma(J) f_{\gamma J}, \quad U_\gamma^{-1} f_J = \varepsilon_{\gamma^{-1}}(J) f_{\gamma^{-1} J},$$

with signs $\varepsilon_\gamma(J) \in \{-1, 1\}$. Define the Reynolds average

$$R_{N,B} = \frac{1}{|G_N(B)|} \sum_{\gamma \in G_N(B)} U_\gamma^{-1} P_N^B R P_N^B U_\gamma.$$

Then $\|R_{N,B}\| \leq C \|R\|$ and $R_{N,B}$ maps $V_N(B)$ into itself while annihilating the complementary Haar levels. For local nodes J, L of levels at most N ,

$$(2.1) \quad \langle f_L^*, R_{N,B} f_J \rangle = \frac{1}{|G_N(B)|} \sum_{\gamma \in G_N(B)} \varepsilon_\gamma(J) \varepsilon_\gamma(L) \langle f_{\gamma L}^*, R f_{\gamma J} \rangle.$$

If $J \neq L$, there is an involution $\rho \in G_N(B)$ fixing the two supports J and L as sets and changing the sign of exactly one of the two Haar functions. Indeed, if J and L are disjoint, flip the two children of one of them; if, say, $L \subsetneq J$, flip the two children of L . In the comparable case the ancestor Haar function is constant on the smaller support, so this smaller child-flip leaves the ancestor Haar function unchanged. Right multiplication by ρ pairs the terms in (2.1) with opposite signs, and therefore

$$\langle f_L^*, R_{N,B} f_J \rangle = 0 \quad (J \neq L).$$

For $J = L$ of relative level k , the sign square is one and the orbit of J under $G_N(B)$ is all of $\mathcal{D}_k(B)$, each node occurring equally often. Hence

$$\langle f_J^*, R_{N,B} f_J \rangle = \frac{1}{\#\mathcal{D}_k(B)} \sum_{M \in \mathcal{D}_k(B)} \langle f_M^*, R f_M \rangle = \beta_k^B(R).$$

Since (f_J, f_J^*) are biorthogonal on the local Haar span, this proves that $R_{N,B}$ is exactly the finite Haar multiplier with coefficient $\beta_k^B(R)$ on all nodes $J \in \mathcal{D}_k(B)$, $0 \leq k \leq N$, and coefficient zero on all other Haar nodes.

Identify the local cancellative space on B with $L_1^0[0, 1]$ by rescaling. The preceding computation gives, on this local space, a finite Haar multiplier whose coefficients are $\beta_k^B(R)$ on the relative levels $0 \leq k \leq N$ and 0 on all deeper relative levels. Applying Theorem 2.1 to this local multiplier, and using $\|R_{N,B}\| \leq C \|R\|$, gives the branch estimate. On a branch below B the variation contains the terminal drop to zero, and hence

$$|\beta_0^B(R)| + \sum_{k=0}^{N-1} |\beta_{k+1}^B(R) - \beta_k^B(R)| + |\beta_N^B(R)| \leq C \|R\|.$$

The constant is independent of B and N . Therefore the finite initial variations of the scalar sequence $(\beta_n^B(R))_n$ are uniformly bounded. Hence

$$\sum_{k=0}^{\infty} |\beta_{k+1}^B(R) - \beta_k^B(R)| < \infty,$$

so $(\beta_n^B(R))_n$ is Cauchy. Its limit $\lambda_B(R)$ exists and satisfies $|\lambda_B(R)| \leq C \|R\|$.

If B_0, B_1 are the dyadic children of B , then for every n

$$\beta_{n+1}^B(R) = \frac{1}{2} \beta_n^{B_0}(R) + \frac{1}{2} \beta_n^{B_1}(R).$$

Passing to limits gives the dyadic additivity identity

$$|B| \lambda_B(R) = |B_0| \lambda_{B_0}(R) + |B_1| \lambda_{B_1}(R).$$

For each dyadic level m put

$$\varphi_m = \sum_{B \in \mathcal{D}_m} \lambda_B(R) \mathbf{1}_B.$$

The preceding identity says that $(\varphi_m)_m$ is an L_∞ -bounded dyadic martingale, and the bound above gives $\|\varphi_m\|_\infty \leq C \|R\|$. By weak-* compactness of bounded subsets of $L_\infty = (L_1)^*$, choose a weak-* cluster point φ_R . If B is dyadic and m is at least the level of B , then

$$\int_B \varphi_m = |B| \lambda_B(R).$$

Passing to the weak-* limit gives

$$\lambda_B(R) = |B|^{-1} \int_B \varphi_R$$

for every dyadic interval B , with $\|\varphi_R\|_\infty \leq C \|R\|$.

Finally, if B is a finite equal-dyadic atom, write

$$B = \bigsqcup_{\ell=1}^r B_\ell$$

as a disjoint union of dyadic intervals all having the same length. Then the relative descendants of B are the union of the relative descendants of the B_ℓ , and hence

$$\beta_n^B(R) = \frac{1}{r} \sum_{\ell=1}^r \beta_n^{B_\ell}(R).$$

Passing to limits and using the dyadic interval formula gives

$$\lambda_B(R) = \frac{1}{r} \sum_{\ell=1}^r \lambda_{B_\ell}(R) = \frac{1}{|B|} \sum_{\ell=1}^r \int_{B_\ell} \varphi_R = \frac{1}{|B|} \int_B \varphi_R.$$

Thus the same trace formula holds for finite equal-dyadic atoms. $\square$

Theorem 2.3 (Semenov–Uksusov stabilization on L_1). *The limits and L_∞ representatives used below are those of Proposition 2.2. In particular,*

$$|\lambda_B(R)| \leq C \|R\|$$

for all $R \in \mathcal{L}(L_1^0)$ and all finite equal-dyadic atoms B .

Proof. This is only a reference label for Proposition 2.2, which was derived from the Semenov–Uksusov branch-variation theorem. $\square$

Theorem 2.4 (LMMS multiplier-to-scalar theorem). *Let D be a bounded bi-parameter Haar multiplier on X_{00} . For every $\eta > 0$ there are operators $A_D, B_D \in \mathcal{L}(X_{00})$ and a scalar λ such that*

$$A_D B_D = \text{Id}_{X_{00}}, \quad \|A_D D B_D - \lambda \text{Id}_{X_{00}}\| < \eta,$$

and $\|A_D\| \|B_D\| \leq 1 + \eta$.

Source of the theorem. This is the scalar corollary of the multiplier theorem of Lechner–Motakis–Muller–Schlumprecht. Their model space is the completion of the linear span $V(\delta^2)$ of the cancellative tensor Haar functions $h_I \otimes h_J$. This is our doubly cancellative product-Haar space X_{00} ; replacing $h_I \otimes h_J$ by the normalized vectors $e_I \otimes f_J$ is only a diagonal change of basis, and conjugates a Haar multiplier to a Haar multiplier with the same coefficient sequence.

In the notation of [1], [1, Theorem 2.14(i)] gives approximate 1-projectional reduction of a bounded Haar multiplier to

$$\lambda \mathcal{C} + \mu(\text{Id} - \mathcal{C}),$$

where $\mathcal{C}$ is Capon’s projection. By [1, Theorem 2.10], Capon’s projection is unbounded on $L_p(L_1)$, $1 < p < \infty$. In the unbounded-Capon case the LMMS conclusion forces $\lambda = \mu$; hence the reduced operator is a scalar multiple of the identity. Equivalently, this is the scalar reduction stated

for $Z = L_p(L_1)$ in [1, Theorem 2.3]. Therefore bounded bi-parameter Haar multipliers on X_{00} approximately 1-projectionally reduce to scalar operators. $\square$

Definition 2.5 (Locked faithful atom trees). A locked faithful atom tree is a family $(\Gamma_I)_{I \in \mathcal{D}}$ of dyadic measurable sets such that $\Gamma_{[0,1]} = [0,1]$, $|\Gamma_I| = |I|$, and for every $I \in \mathcal{D}$, the sets Γ_{I+} and Γ_{I-} partition Γ_I into two sets of equal measure. In the constructions below every atom, outer and inner, is a finite equal-dyadic atom: a finite disjoint union of dyadic intervals of one common length. This class is preserved by the packet splits used in Section 3 and is the class for which the local trace theorem is applied. The associated normalized Haar blocks are

$$H_I = |\Gamma_I|^{-1/p}(\mathbf{1}_{\Gamma_{I+}} - \mathbf{1}_{\Gamma_{I-}}), \quad H_I^* = |\Gamma_I|^{-1/p'}(\mathbf{1}_{\Gamma_{I+}} - \mathbf{1}_{\Gamma_{I-}})$$

in the outer coordinate, and

$$K_J = |\Delta_J|^{-1}(\mathbf{1}_{\Delta_{J+}} - \mathbf{1}_{\Delta_{J-}}), \quad K_J^* = \mathbf{1}_{\Delta_{J+}} - \mathbf{1}_{\Delta_{J-}}$$

in the inner coordinate.

Lemma 2.6 (Exterior maps for locked faithful systems). *Let (Γ_I) and (Δ_J) be locked faithful atom trees. The maps determined on the product Haar span by*

$$Bu_{I,J} = H_I \otimes K_J,$$

and

$A(H_I \otimes K_J) = u_{I,J}$, $Az = 0$ on the coefficient-annihilator of the faithful product algebra, extend to bounded operators on X_{00} satisfying

$$AB = \text{Id}_{X_{00}}, \quad \|A\| \leq 1, \quad \|B\| \leq 1.$$

Moreover,

$$\langle u_{I,J}^*, ATBu_{I',J'} \rangle = \langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_{J'}) \rangle$$

for every bounded operator T for which the expression is defined.

Proof. The atom tree (Γ_I) defines a measure algebra isomorphism from the standard dyadic algebra onto the algebra generated by the sets Γ_I ; similarly for (Δ_J) . Pullback by the product of these two measure algebra embeddings is an isometry on $L_p(L_1)$ and sends $u_{I,J}$ to $H_I \otimes K_J$. This is B .

The map A is the conditional expectation onto the faithful product σ -algebra, followed by the inverse measure-algebra map. Conditional expectation is contractive on $L_p(L_1)$ because it is contractive on L_1 in the inner variable and on L_p in the outer variable. It preserves the zero-integral condition in each coordinate: if, for instance, $\mathbb{E}_2 z = 0$, then the inner conditional expectation still has integral zero on each outer atom, and similarly in the outer coordinate. Hence A maps the faithful product-cancellative part back into X_{00} . Thus $\|A\| \leq 1$, $\|B\| \leq 1$, and $AB = \text{Id}$. The coefficient identity follows from the definitions of the biorthogonal blocks. $\square$

3. LOCAL TRACES AND SHADOW COMPACTNESS

Let A be an outer atom, i.e. one terminal set in a finite locked faithful outer tree. For $m \geq 0$, let $\mathcal{D}_m(A)$ be the equal-measure dyadic descendants of A at relative depth m . Each $I \in \mathcal{D}_m(A)$ has its own normalized Haar vector e_I and functional e_I^* as above. For $Q \in \mathcal{L}(L_p^0)$ define

$$\alpha_m^A(Q) = \frac{1}{\#\mathcal{D}_m(A)} \sum_{I \in \mathcal{D}_m(A)} \langle e_I^*, Qe_I \rangle.$$

For an inner atom B and $R \in \mathcal{L}(L_1^0)$, define $\beta_n^B(R)$ similarly, as in Theorem 2.3.

Definition 3.1 (Common tails). Let A be an outer atom, let $\mathcal{Q} \subset \mathcal{L}(L_p^0)$ be finite, and let $\rho > 0$. An infinite set $S_A \subset \mathbb{N}$ is a common ρ -tail for $(A, \mathcal{Q})$ if

$$|\alpha_m^A(Q)| < \rho \quad (Q \in \mathcal{Q}, m \in S_A).$$

The inner definition uses $\beta_n^B(R)$ in place of $\alpha_m^A(Q)$.

For $T \in \mathcal{L}(X_{00})$, an outer atom A , and $g \in L_1^0$, define

$$\Omega_{A,m}^T g = \frac{1}{\#\mathcal{D}_m(A)} \sum_{I \in \mathcal{D}_m(A)} (e_I^* \otimes \text{Id}) T(e_I \otimes g) \in L_1^0.$$

Equivalently,

$$\langle \phi, \Omega_{A,m}^T g \rangle = \frac{1}{\#\mathcal{D}_m(A)} \sum_{I \in \mathcal{D}_m(A)} \langle e_I^* \otimes \phi, T(e_I \otimes g) \rangle, \quad \phi \in L_\infty.$$

Proposition 3.2 (Outer shadows are L_1 -valued). *Let $T \in \mathcal{L}(X_{00})$, let A be an outer atom, let $S \subset \mathbb{N}$ be infinite, and let $g \in L_1^0$. Then*

$$\{\Omega_{A,m}^T g : m \in S\}$$

is uniformly integrable in L_1 . Consequently, if $\mathcal{U}$ is a free ultrafilter containing S , then

$$\Omega_A^{T,\mathcal{U}} g = \text{weak} \lim_{m \rightarrow \mathcal{U}} \Omega_{A,m}^T g$$

exists as an element of L_1^0 , and $\Omega_A^{T,\mathcal{U}} : L_1^0 \rightarrow L_1^0$ is bounded with

$$\|\Omega_A^{T,\mathcal{U}}\| \leq C_p \|T\|.$$

Proof. First the orbit is bounded in L_1 . Let $N_m = \#\mathcal{D}_m(A)$ and, for signs $\sigma = (\sigma_I)$, put

$$x_m^\sigma = N_m^{-1/p} \sum_{I \in \mathcal{D}_m(A)} \sigma_I e_I, \quad (x_m^\sigma)^* = N_m^{-1/p'} \sum_{I \in \mathcal{D}_m(A)} \sigma_I e_I^*.$$

Then $\|x_m^\sigma\|_p = \|(x_m^\sigma)^*\|_{p'} = 1$, and averaging over σ gives

$$\langle \phi, \Omega_{A,m}^T g \rangle = \mathbb{E}_\sigma \langle (x_m^\sigma)^* \otimes \phi, T(x_m^\sigma \otimes g) \rangle.$$

Thus $\|\Omega_{A,m}^T g\|_1 \leq \|T\| \|g\|_1$.

Suppose uniform integrability fails for some g with $\|g\|_1 = 1$. By the Kadec–Pelczynski disjointification criterion in L_1 , after passing to a subsequence $m_1 < m_2 < \dots$ in S , there are pairwise disjoint measurable inner sets E_k and a number $\delta > 0$ such that

$$\int_{E_k} |\Omega_{A,m_k}^T g| > \delta \quad (k \geq 1).$$

Choose $\psi_k \in L_\infty$, supported on E_k , with $\|\psi_k\|_\infty \leq 1$ and

$$\langle \psi_k, \Omega_{A,m_k}^T g \rangle > \delta.$$

For each k , let $x_k^{\sigma_k} = x_{m_k}^{\sigma_k}$. Then

$$\mathbb{E}_{\sigma_k} \langle (x_k^{\sigma_k})^* \otimes \psi_k, T(x_k^{\sigma_k} \otimes g) \rangle > \delta.$$

Introduce independent signs $\varepsilon_1, \dots, \varepsilon_N$. Averaging over these signs kills all cross terms, hence

$$\delta N < \mathbb{E}_{\sigma, \varepsilon} \left\langle \sum_{k=1}^N \varepsilon_k (x_k^{\sigma_k})^* \otimes \psi_k, T \left(\sum_{k=1}^N \varepsilon_k x_k^{\sigma_k} \otimes g \right) \right\rangle.$$

The dual vector has norm at most 1 in $L_{p'}(L_\infty)$. Indeed, because the supports of the ψ_k are disjoint in the inner variable,

$$\left\| \sum_{k=1}^N \varepsilon_k (x_k^{\sigma_k})^*(s) \psi_k \right\|_\infty \leq |A|^{-1/p'} \mathbf{1}_A(s)$$

for every outer point s , and the right hand side has $L_{p'}$ norm one. Also,

$$\mathbb{E}_{\sigma, \varepsilon} \left\| \sum_{k=1}^N \varepsilon_k x_k^{\sigma_k} \right\|_{L_p} \leq C_p N^{1/2},$$

by the scalar Khintchine inequality pointwise on A . Therefore $\delta N \leq C_p \|T\| N^{1/2}$ for all N , a contradiction. The orbit is uniformly integrable.

By Dunford–Pettis, a bounded uniformly integrable set in L_1 is relatively weakly compact. Thus every ultrafilter on S has a weak limit in L_1 . Since $L_1^0 = \ker \mathbb{E}$ is weakly closed, this limit belongs to L_1^0 . Scalar ultralimits show linearity in g , and the initial uniform bound gives the stated operator norm estimate. $\square$

For an inner atom B , define first a bilinear form

$$\mathbf{i}_B^T(x^*, x) = \lim_{n \rightarrow \infty} \beta_n^B((x^* \otimes \text{Id})T(x \otimes \cdot)),$$

for $x \in L_p^0$ and $x^* \in L_{p'}^0$. The limit exists by Theorem 2.3, and

$$|\mathbf{i}_B^T(x^*, x)| \leq C \|T\| \|x^*\|_{p'} \|x\|_p.$$

Since $1 < p < \infty$, the space L_p^0 is reflexive and $(L_{p'}^0)^*$ is canonically represented by L_p^0 after quotienting constants and taking the zero-mean representative. Thus there is a unique operator $\mathcal{I}_B^T \in \mathcal{L}(L_p^0)$ satisfying

$$\langle x^*, \mathcal{I}_B^T x \rangle = \mathbf{i}_B^T(x^*, x), \quad x \in L_p^0, \quad x^* \in L_{p'}^0,$$

and $\|\mathcal{I}_B^T\| \leq C \|T\|$.

4. FINITE SPLITTING ESTIMATES

Write

$$L_p^0 = \overline{\text{span}}\{h_I : I \in \mathcal{D}\} \subset L_p[0, 1], \quad L_1^0 = \overline{\text{span}}\{h_J : J \in \mathcal{D}\} \subset L_1[0, 1].$$

When functionals on these zero-mean subspaces are represented by functions, we use the standard quotient convention: $(L_p^0)^*$ is represented by $L_{p'}$ modulo constants and $(L_1^0)^*$ by L_∞ modulo constants. We always choose a representative whose norm is within a fixed absolute factor of the quotient norm; adding a constant never changes any pairing with a zero-mean Haar block. This convention is used silently in the finite side-constraint estimates.

The following estimates are used only for finite families. Constants may depend on p and on the number of admissible side constraints, but not on the chosen splitting depths.

The random split used in this section is always the following packet split. A current atom is first refined into N equal dyadic packets $L_1, \dots, L_N$; each packet is then split into its two dyadic children. Independent signs $\varepsilon_i \in \{-1, 1\}$ decide which child of L_i goes to the plus child of the current atom and which goes to the minus child. Thus the two children have equal measure automatically, and the resulting Haar block is the normalized signed average of the packet Haar vectors. All probability estimates below are with respect to these independent packet signs.

Definition 4.1 (Admissible finite side constraints). Let E be either L_p^0 or L_1^0 , and suppose that a finite family of current terminal atoms $C_1, \dots, C_d$ is being split. Write u_a, u_a^* for the random normalized block and its biorthogonal functional born on C_a . A scalar side constraint imposed at this splitting stage is called *admissible* if, after all older blocks have been frozen, its variable part is one of the following forms:

$$|\langle u_a^*, Sz \rangle| < \eta, \quad | \langle z^*, Su_a \rangle | < \eta, \quad | \langle u_a^*, Su_b \rangle | < \eta \quad (a \neq b),$$

where $S \in \mathcal{L}(E)$ is fixed for the stage, $z \in E$ and $z^* \in E^*$ are already constructed fixed data, and $\eta > 0$ is the prescribed tolerance. The self-diagonal expression $\langle u_a^*, Su_a \rangle$ is not a side constraint; it is allowed only when it is one of the separately treated trace/diagonal estimates below. Product-space coefficient constraints used in the locked construction are admissible precisely when, after freezing the inactive coordinate and all older blocks, they reduce in the active coordinate to one of the three displayed forms.

Definition 4.2 (Locally finite admissible schedules). A countable family of admissible side constraints in a recursive construction is called *locally finite* if each constraint is assigned to the unique half-step at which its newest active block is born, and only finitely many constraints are assigned to each half-step. All countable side schedules used below are understood in this sense. Thus every finite splitting lemma is applied only to a finite list of constraints, while the countable family is exhausted over the recursion.

Lemma 4.3 (Centered quadratic estimate). Let $E = L_p^0$ or $E = L_1^0$, let $Q \in \mathcal{L}(E)$, and let u, u^* be the normalized random faithful Haar block obtained by splitting an atom into N equal packets. If u_i, u_i^* are the normalized packet Haar vectors and functionals, then

$$\mathbb{E} \left| \langle u^*, Qu \rangle - \frac{1}{N} \sum_{i=1}^N \langle u_i^*, Qu_i \rangle \right|^2 \leq \frac{C_p \|Q\|^2}{N}.$$

Proof. Let the packet vectors be supported on pairwise disjoint packets and put

$$q_{ij} = \langle u_i^*, Qu_j \rangle.$$

For $E = L_p^0$ we have

$$u = N^{-1/p} \sum_{i=1}^N \varepsilon_i u_i, \quad u^* = N^{-1/p'} \sum_{i=1}^N \varepsilon_i u_i^*,$$

and for $E = L_1^0$ the same formula is read as

$$u = N^{-1} \sum_{i=1}^N \varepsilon_i u_i, \quad u^* = \sum_{i=1}^N \varepsilon_i u_i^*.$$

In both cases

$$\langle u^*, Qu \rangle - \frac{1}{N} \sum_i q_{ii} = \frac{1}{N} \sum_{i \neq j} \varepsilon_i \varepsilon_j q_{ij}.$$

Independence of the signs gives, with a harmless factor accounting for the symmetric pairs (i, j) and (j, i) ,

$$\mathbb{E} \left| \frac{1}{N} \sum_{i \neq j} \varepsilon_i \varepsilon_j q_{ij} \right|^2 \leq 2N^{-2} \sum_{i \neq j} |q_{ij}|^2.$$

It remains to bound this Hilbert–Schmidt sum. Let P be the coordinate projection onto the disjoint packet span. It is the relevant finite martingale-difference projection, so $\|P\| \leq C_p$ on L_p and on L_1 . For a fixed column j , the vector of coefficients $(q_{ij})_i$ is the coordinate vector of

PQu_j . Hence, in L_1 , its ℓ_1 -norm is at most $C\|Q\|$, and in L_p with $1 < p \leq 2$ its ℓ_p -norm is at most $C_p\|Q\|$. In both cases its ℓ_2 -norm is at most $C_p\|Q\|$. Summing over the N columns gives $\sum_{i,j} |q_{ij}|^2 \leq C_p N \|Q\|^2$.

For $p \geq 2$, apply the same argument to the rows, equivalently to P^*Q^* on $L_{p'}$ with $p' \leq 2$. This again gives $\sum_{i,j} |q_{ij}|^2 \leq C_p N \|Q\|^2$. Substitution in the preceding display proves the estimate. $\square$

Lemma 4.4 (One-sided estimates). *Let $E = L_p^0$ or $E = L_1^0$. Let u, u^* be a random faithful Haar block obtained by splitting one atom into N equal packets. For every finite family of fixed vectors $z_b \in E$ and fixed functionals $z_a^* \in E^*$, the depth can be chosen so large that*

$$\mathbb{E} |\langle u^*, z_b \rangle|^2 < \eta^2, \quad \mathbb{E} |\langle z_a^*, u \rangle|^2 < \eta^2$$

for all a, b . The same conclusion holds after replacing z_b by Sz_b and z_a^* by $S^*z_a^*$ for any fixed finite family of bounded operators S .

Proof. For L_1 , write the packets as L . If $z \in L_1$, then

$$\mathbb{E} |\langle u^*, z \rangle|^2 = \sum_L \left| \int_L f_L^* z \right|^2 \leq \left(\max_L \int_L |z| \right) \|z\|_1,$$

which tends to zero as the packets are made sufficiently fine. If $z^* \in L_\infty$, then

$$\mathbb{E} |\langle z^*, u \rangle|^2 = N^{-2} \sum_L |\langle z^*, f_L \rangle|^2 \leq N^{-1} \|z^*\|_\infty^2.$$

For L_p , let $u = N^{-1/p} \sum_L \varepsilon_L u_L$ and $u^* = N^{-1/p'} \sum_L \varepsilon_L u_L^*$. If $z \in L_p$, then

$$\mathbb{E} |\langle u^*, z \rangle|^2 = N^{-2/p'} \sum_L |\langle u_L^*, z \rangle|^2, \quad |\langle u_L^*, z \rangle| \leq \|\mathbf{1}_L z\|_p.$$

The last display tends to zero under refinement. For $p \leq 2$ use

$$\sum_L \|\mathbf{1}_L z\|_p^2 \leq (\max_L \|\mathbf{1}_L z\|_p)^{2-p} \|z\|_p^p,$$

while for $p > 2$ Hölder's inequality gives an additional factor N^{-1} after the normalization. The estimate for $\mathbb{E} |\langle z^*, u \rangle|^2$ is the same argument in $L_{p'}$. Applying these estimates to the finite family Sz_b and $S^*z_a^*$ gives the operator version. $\square$

Lemma 4.5 (Bilinear estimates for two new blocks). *Let $E = L_p^0$ or $E = L_1^0$, let $S \in \mathcal{L}(E)$, and let u_a, u_a^* and u_b, u_b^* be independent random faithful blocks built on two disjoint atoms, each with at least N packets. Then*

$$\mathbb{E} |\langle u_a^*, Su_b \rangle|^2 \leq \frac{C_p \|S\|^2}{N}.$$

Proof. Let the two packet families have cardinalities $N_a, N_b \geq N$, and write

$$s_{ij} = \langle u_{a,i}^*, Su_{b,j} \rangle.$$

The signs in the two packet families are independent, so the second moment of $\langle u_a^*, Su_b \rangle$ is the squared normalization times $\sum_{i,j} |s_{ij}|^2$.

In the L_1 case the block vector has normalization N_b^{-1} and the L_∞ block functional has no cardinal normalization. Hence

$$\mathbb{E} |\langle u_a^*, Su_b \rangle|^2 = N_b^{-2} \sum_{i,j} |s_{ij}|^2.$$

For each fixed j , the column $(s_{ij})_i$ is the packet-coordinate vector of $Su_{b,j}$ in an L_1 disjoint Haar packet span; its ℓ_1 norm is at most $C \|S\|$. Therefore its ℓ_2 norm is at most $C \|S\|$, and

$$\mathbb{E} |\langle u_a^*, Su_b \rangle|^2 \leq C N_b^{-1} \|S\|^2 \leq C N^{-1} \|S\|^2.$$

For $E = L_p^0$, the normalization is $N_a^{-1/p'} N_b^{-1/p}$. If $1 < p \leq 2$, each column has ℓ_p , hence ℓ_2 , norm at most $C_p \|S\|$. Thus

$$\mathbb{E} |\langle u_a^*, Su_b \rangle|^2 \leq C_p N_a^{-2/p'} N_b^{-2/p} N_b \|S\|^2 = C_p N_a^{-2/p'} N_b^{1-2/p} \|S\|^2 \leq C_p N^{-1} \|S\|^2.$$

If $p \geq 2$, apply the same argument to rows, equivalently to S^* on the dual packet span where $p' \leq 2$. This gives

$$\mathbb{E} |\langle u_a^*, Su_b \rangle|^2 \leq C_p N_a^{1-2/p'} N_b^{-2/p} \|S\|^2 \leq C_p N^{-1} \|S\|^2.$$

The estimate follows in all cases. $\square$

5. ONE-PARAMETER STRIP EXTENSION LEMMAS

The locked construction estimates strips, not individual coefficients in one absolutely summable table. We need one extension lemma in the outer L_p coordinate and one in the inner L_1 coordinate.

5.1. The outer L_p strip extension. Let A be an outer terminal atom and $Q \in \mathcal{L}(L_p^0)$. Recall

$$\alpha_m^A(Q) = \frac{1}{\#\mathcal{D}_m(A)} \sum_{I \in \mathcal{D}_m(A)} \langle e_I^*, Q e_I \rangle.$$

An infinite set $S_A \subset \mathbb{N}$ is a common (θ_a) -tail for a finite family (Q_a) if

$$|\alpha_m^A(Q_a)| < \theta_a \quad (m \in S_A, a \in F).$$

Lemma 5.1 (Additive preservation of outer tails). *Let $A_1, \dots, A_d$ be outer terminal atoms. For each a , let $(Q_s)_{s \in F_a} \subset \mathcal{L}(L_p^0)$ be finite, let $S_a \subset \mathbb{N}$ be an infinite common $(\theta_s)_{s \in F_a}$ -tail, and let $(\delta_s)_{s \in F_a}$ be positive. Let $\mathcal{S}$ be a finite list of admissible side constraints, in the sense of Definition 4.1, involving the new outer blocks. Then there are splitting depths $n_a \in S_a$ and faithful splits of all A_a such that:*

(i) *the new block H_{A_a} satisfies*

$$|\langle H_{A_a}^*, Q_s H_{A_a} \rangle| < \theta_s + \delta_s \quad (s \in F_a);$$

(ii) *all admissible side constraints in $\mathcal{S}$ have modulus below their prescribed tolerances;*

(iii) *each child of each A_a has an infinite common $(\theta_s + \delta_s)_{s \in F_a}$ -tail for the same family.*

Proof. Choose the depths $n_a \in S_a$ so large that the packet numbers make all variances in Lemmas 4.3, 4.4, and 4.5 as small as needed below. Let E be the event that all diagonal estimates in (i) and all admissible side constraints in $\mathcal{S}$ hold. By Definition 4.1, those admissible side constraints are exactly of the forms covered by Lemmas 4.4 and 4.5. Since the expectations of the new diagonal coefficients are the parent traces $\alpha_{n_a}^{A_a}(Q_s)$, whose moduli are $< \theta_s$, Chebyshev's inequality and the finite union bound let us arrange

$$(1) \quad \mathbb{P}(E) > 1 - \eta$$

for any prescribed $\eta > 0$.

It remains to arrange infinite child tails without losing the event E . Fix an atom A_a . Enumerate the eligible future depths by

$$r_1 < r_2 < \dots, \quad n_a + 1 + r_\ell \in S_a.$$

For an eligible depth r_ℓ , let $H_{a,\ell}$ be the event that, for both children $A_{a,+}$, $A_{a,-}$ and for every $s \in F_a$, the trace at relative depth r_ℓ below that child has modulus $< \theta_s + \delta_s$. To see the probability

estimate, write the split at depth n_a packetwise: from each packet one dyadic child is sent to $A_{a,+}$ and the other to $A_{a,-}$. The trace below, say, $A_{a,+}$ at relative depth r_ℓ is the average of the N_a independent random variables obtained by choosing one of the two packet-child descendant averages. Each variable is bounded by $\|Q_s\|$, and its expectation is the parent trace at absolute depth $n_a + 1 + r_\ell$, hence has modulus $< \theta_s$. Therefore the variance is at most $C_p N_a^{-1} \|Q_s\|^2$, uniformly in ℓ ; the same argument applies to $A_{a,-}$. Increasing n_a if necessary, we may therefore make

$$\mathbb{P}(H_{a,\ell}^c) < \kappa \quad (\ell \geq 1, 1 \leq a \leq d)$$

with $\kappa > 0$ as small as desired.

For $M \geq 1$, let $F_{a,M}$ be the event that at least M of $H_{a,1}, \dots, H_{a,2M}$ occur. The expected number of failures among these $2M$ events is at most $2M\kappa$, and Markov's inequality gives

$$\mathbb{P}(F_{a,M}^c) \leq 2\kappa.$$

Consequently

$$\mathbb{P}\left(E \cap \bigcap_{a=1}^d F_{a,M}\right) \geq 1 - \eta - 2d\kappa.$$

Choose η and κ so that the right-hand side is positive. Thus, for every M , there is a single simultaneous sign choice satisfying E and giving at least M good eligible depths below every child of every atom. The sign space for the already fixed finite splitting depths is finite. Hence one sign choice occurs for infinitely many values of M . That sign choice satisfies E and has infinitely many good eligible depths below each child. These infinite good-depth sets are the required child common tails. $\square$

Proposition 5.2 (Small outer multiplier strips). *Let $(Q_s)_{s \geq 1}$ be operators on L_p^0 which are introduced during a recursive construction, and let (σ_s) be summable. Assume that the construction maintains the following common-tail invariant: at each finite outer stage and for every current outer terminal atom A , there is an infinite common tail S_A for the finite family of active same-outer strips below A , and whenever a new Q_s is introduced this common tail is refined so that*

$$|\alpha_m^A(Q_s)| < \sigma_s/4 \quad (m \in S_A)$$

for every current terminal atom A . Then the outer faithful tree can be continued so that all future diagonal coefficients

$$a_I^{(s)} = \langle H_I^*, Q_s H_I \rangle$$

satisfy $|a_I^{(s)}| < \sigma_s/2$ after the birth of Q_s , while every prescribed locally finite schedule of admissible side constraints is also satisfied with prescribed summable tolerances. Consequently the descendant part of the corresponding outer Haar multipliers has norm at most $C_p \sigma_s$. If the finitely many pre-birth coefficients have separately been bounded by $\sigma_s/2$, then the full strip multiplier has the same bound, after increasing C_p . Thus the sum of such strips has norm at most $C_p \sum_s \sigma_s$.

Proof. At the N -th outer splitting stage only finitely many Q_s are active. For a fixed current terminal atom A , the invariant supplies one common tail S_A on which all active trace bounds hold with their current allowances. Apply Lemma 5.1 to the finite active family on these common tails, assigning to Q_s an increment sequence whose total after the birth of Q_s is $< \sigma_s/4$. The lemma returns child common tails for the same active family. Thus the common-tail invariant is preserved, and the actual future diagonal coefficients remain below $\sigma_s/2$.

The locally finite admissible side schedule is handled in the same finite-stage manner: by Definition 4.2, only finitely many constraints are assigned to the current half-step. Include exactly those constraints in the application of Lemma 5.1. This keeps every application finite and imposes each admissible side constraint exactly once.

The scalar Haar multiplier on L_p , $1 < p < \infty$, has norm at most the unconditional constant of the Haar system times the supremum of its coefficients. The same estimate applies to a finite pre-birth part once those finitely many coefficients have been imposed at their registration time as admissible side constraints and bounded by $\sigma_s/2$. Summability of (σ_s) gives the final assertion. $\square$

5.2. The inner L_1 strip extension. For $R \in \mathcal{L}(L_1^0)$ and an inner atom B , put

$$\lambda_B(R) = \lim_{n \rightarrow \infty} \beta_n^B(R),$$

where the limit exists by Theorem 2.3.

Lemma 5.3 (The local L_1 trace is an L_∞ average). *For every $R \in \mathcal{L}(L_1^0)$ there is $\varphi_R \in L_\infty[0, 1]$ with $\|\varphi_R\|_\infty \leq C \|R\|$ such that, for every dyadic interval or finite equal-dyadic atom B ,*

$$\lambda_B(R) = \frac{1}{|B|} \int_B \varphi_R.$$

Proof. This is Proposition 2.2. $\square$

Lemma 5.4 (One-step L_1 split with trace control). *Let $R_1, \dots, R_m \in \mathcal{L}(L_1^0)$, let B be an inner terminal atom, and let $\delta > 0$. Let $\mathcal{S}$ be a finite list of admissible side constraints involving the new inner block K_B, K_B^* and already fixed data, each with its own prescribed tolerance. Then for every $\gamma > 0$, all sufficiently large equal-packet splitting depths have the following property: the random faithful split of B satisfies, with probability at least $1 - \gamma$, for every $r \leq m$,*

$$|\langle K_B^*, R_r K_B \rangle - \lambda_B(R_r)| < \delta,$$

$$|\lambda_{B_+}(R_r) - \lambda_B(R_r)| < \delta, \quad |\lambda_{B_-}(R_r) - \lambda_B(R_r)| < \delta,$$

and every admissible side constraint in $\mathcal{S}$ has modulus below its prescribed tolerance. In particular, such a faithful split exists.

Proof. Choose the relative depth n so large that $|\beta_n^B(R_r) - \lambda_B(R_r)| < \delta/4$ for every r . The expectation of the new diagonal coefficient is $\beta_n^B(R_r)$, and Lemma 4.3 gives variance $O(N^{-1} \|R_r\|^2)$. After increasing the number N of packets, Chebyshev's inequality makes the probability of failure of all diagonal estimates smaller than any prescribed number.

Let $\varphi_r = \varphi_{R_r}$ be given by Lemma 5.3. For each packet L with two children L_0, L_1 , the average of φ_r over a randomly selected child has expectation equal to the average of φ_r over L and differs from it by a signed number bounded by $2 \|\varphi_r\|_\infty$. Averaging over all packets gives variance $O(N^{-1})$ for the plus child; the minus child is handled identically, or by the identity $\lambda_B(R_r) = \frac{1}{2} \lambda_{B_+}(R_r) + \frac{1}{2} \lambda_{B_-}(R_r)$. Hence the two child-trace estimates also hold with arbitrarily high probability. The finite admissible side constraints are handled by Lemmas 4.4 and 4.5; in the one-atom case only the one-sided forms can involve the newly born block, while bilinear constraints are reserved for simultaneous splittings on distinct atoms. A finite union bound gives probability at least $1 - \gamma$. $\square$

Lemma 5.5 (Simultaneous finite L_1 splitting). *Let $B_1, \dots, B_d$ be inner terminal atoms. For each a , let $\mathcal{R}_a$ be a finite family of operators on L_1^0 , and let positive tolerances be prescribed for the diagonal estimates and child-trace estimates associated with those operators. Let $\mathcal{S}$ be a finite list of admissible side constraints for the simultaneous splitting of $B_1, \dots, B_d$. Thus every member of $\mathcal{S}$ is one-sided with one newly born block and fixed old data, or bilinear between two newly born blocks supported on distinct current terminal atoms. Then, after taking sufficiently deep equal-packet refinements of the atoms, there are faithful splits of all B_a for which all the prescribed trace estimates and all constraints in $\mathcal{S}$ hold simultaneously.*

Proof. Use the high-probability form of Lemma 5.4, and choose the refinement depths so that the one-atom trace estimates have failure probability smaller than a prescribed number after a union bound over a and over all operators in $\mathcal{R}_a$. The one-sided admissible constraints are controlled by Lemma 4.4. A bilinear admissible constraint necessarily involves two distinct current atoms, and is controlled by Lemma 4.5, since the atoms are disjoint and their sign choices are independent. Self-diagonal constraints are not in $\mathcal{S}$; the only expressions of the form $\langle K_B^*, RK_B \rangle$ are the separately prescribed trace estimates. Since the list of constraints is finite, all failure probabilities can be made to have total less than one. Hence a simultaneous sign choice satisfying every requested estimate exists. $\square$

For a scalar sequence (d_C) indexed by the nodes of a full descendant dyadic forest, i.e. a finite collection of roots together with all of their descendants, define

$$\text{TV}_{\text{full}}(d) = \sum_{R \in \mathcal{R}} |d_R| + \sum_{C \rightarrow C'} |d_C - d_{C'}|,$$

where $\mathcal{R}$ is the root set and the second sum runs over all forest edges. The word “full” is important: after a branch enters the forest at a root, it never leaves the forest. For a partial finite forest one must add the boundary term $\sum_{C \in \partial \mathcal{F}} |d_C|$ for terminal leaves; we shall not use that partial version below. The Semenov–Uksusov norm uses the weaker branch variation

$$\text{BV}_{\text{br}}(d) = \sup_{C_0 \supset C_1 \supset \dots} \left(|d_{C_0}| + \sum_k |d_{C_{k+1}} - d_{C_k}| + \limsup_k |d_{C_k}| \right).$$

For each branch,

$$\limsup_k |d_{C_k}| \leq |d_{C_0}| + \sum_k |d_{C_{k+1}} - d_{C_k}|.$$

Thus $\text{BV}_{\text{br}}(d) \leq 2 \text{TV}_{\text{full}}(d)$ on a full descendant forest.

Lemma 5.6 (Forest-to-full-tree multiplier estimate). *Let (d_C) be a scalar sequence on a full descendant dyadic forest in a faithful L_1 Haar tree. Let M_d be the corresponding Haar multiplier on the closed span of that forest, extended by zero on all Haar nodes outside the forest after identifying the faithful tree with the standard dyadic tree. Then*

$$(4.1) \quad \|M_d : L_1^0 \rightarrow L_1^0\| \leq C_F \text{TV}_{\text{full}}(d).$$

Proof. Pull the faithful tree back to the standard dyadic tree by the measure-algebra isomorphism associated with the faithful splits. Extend d to a full dyadic coefficient sequence $\tilde{d}$ by setting $\tilde{d}_C = d_C$ on forest nodes and $\tilde{d}_C = 0$ off the forest. If a dyadic branch never enters the forest, its variation is zero. If it first enters the forest at a root R , the only extra jump from the zero extension is $|d_R - 0|$, which is one of the root terms in $\text{TV}_{\text{full}}(d)$. Since the forest is full, the branch then remains inside the forest forever. The remaining branch variation, including the terminal limsup term, is bounded by the corresponding forest branch variation. Hence

$$\text{BV}_{\text{br}}(\tilde{d}) \leq 2 \text{TV}_{\text{full}}(d).$$

The Semenov–Uksusov theorem applied to the full-tree multiplier with coefficients $\tilde{d}$ gives (4.1). Restricting back to the forest span cannot increase the norm. $\square$

Lemma 5.7 (Glued-tree multiplier estimate). *Let a finite old dyadic tree with coefficients (b_P) be glued to full descendant forests rooted at its finite frontier $\mathcal{R}$, so that after the attachment there are no terminal leaves. A frontier root $B \in \mathcal{R}$ either has no old parent, in which case it is a root*

of the glued forest, or has an old parent $p(B)$. Let (d_C) be the coefficients on the frontier roots and all future descendants. If

$$\begin{aligned} \text{TV}_{\text{glue}} = & \mathbf{1}_{\{C_{\text{root}} \text{ is old}\}} |b_{C_{\text{root}}}| + \sum_{P \rightarrow Q : P, Q \text{ old}} |b_P - b_Q| \\ & + \sum_{B \in \mathcal{R} : p(B) \text{ old}} |b_{p(B)} - d_B| + \sum_{B \in \mathcal{R} : p(B) \text{ absent}} |d_B| + \sum_{C \rightarrow C' \text{ future}} |d_C - d_{C'}|, \end{aligned}$$

then the multiplier on the glued tree has norm at most $C_F \text{TV}_{\text{glue}}$.

Proof. The old tree together with the attached full descendant forests is itself a full descendant forest, with roots given by the old root when it is present and by the frontier roots without old parents otherwise. Its full variation is exactly bounded by the displayed glued variation. Apply Lemma 5.6. $\square$

Proposition 5.8 (Small L_1 multiplier strips). *Let $(R_s)_{s \geq 1}$ be operators on L_1^0 introduced during a recursive construction, and let (τ_s) be summable. When R_s is introduced, let $\mathcal{T}_s^{\text{old}}$ be the already built finite inner tree for this strip, with fixed old coefficients $b_P^{(s)}$ on its born nodes, and let $\mathcal{R}_s$ be the current terminal frontier, which is the root set of the future descendant forest of this strip. Define the old charge*

$$\begin{aligned} (4.2a) \quad G_s = & \mathbf{1}_{\{C_{\text{root}} \in \mathcal{T}_s^{\text{old}}\}} |b_{C_{\text{root}}}^{(s)}| + \sum_{P \rightarrow Q : P, Q \in \mathcal{T}_s^{\text{old}}} |b_P^{(s)} - b_Q^{(s)}| \\ & + \sum_{B \in \mathcal{R}_s : p(B) \in \mathcal{T}_s^{\text{old}}} |b_{p(B)}^{(s)}|. \end{aligned}$$

Here $C_{\text{root}} = [0, 1)$, and $p(B)$ denotes the old parent of the terminal atom B , if such a parent exists. The last sum deliberately counts the same old endpoint once for each adjacent birth root. If there is no old inner part, then $G_s = 0$. Assume that

$$(4.2) \quad \sum_{B \in \mathcal{R}_s} |\lambda_B(R_s)| + G_s < \tau_s / C_0,$$

where C_0 is a sufficiently large absolute bookkeeping constant.

Then the inner faithful tree can be continued so that, for each s , the complete glued coefficient sequence of the strip, consisting of the old part, the gluing edges, and the descendant forest born after R_s , has glued variation $< \tau_s$. Every prescribed locally finite schedule of admissible side constraints can be imposed simultaneously. Hence the corresponding inner L_1 multiplier strips have total norm at most $C \sum_s \tau_s$.

Proof. Fix one strip R_s . In this abstract proposition, (4.2) is a hypothesis on the already fixed old coefficients and on the birth-root traces. In Theorem 7.1, this hypothesis is enforced at the registering outer half-step by admissible outer side constraints.

Let $\mathcal{F}_s$ be the future descendant forest below the roots $\mathcal{R}_s$. Choose positive node-error budgets $\eta_C^{(s)}$ for every future node $C \in \mathcal{F}_s$, and positive edge budgets $\rho_{C, C'}^{(s)}$ for every future forest edge $C \rightarrow C'$, so small that

$$(4.3) \quad G_s + 3 \sum_{B \in \mathcal{R}_s} |\lambda_B(R_s)| + 4 \sum_{C \in \mathcal{F}_s} \eta_C^{(s)} + \sum_{C \rightarrow C'} \rho_{C, C'}^{(s)} < \tau_s.$$

This is possible by (4.2), after increasing the absolute constant C_0 . The numerical factors are only multiplicity allowances: a birth-root trace is used in at most one root-or-gluing estimate, and a node-error term is used once in that root-or-gluing estimate or incoming edge and at most twice in the two outgoing edge estimates.

At each finite inner splitting stage only finitely many strips and admissible side constraints are active. We apply Lemma 5.5 to all active R_s on all current terminal atoms. When a future node $C \in \mathcal{F}_s$ is born, write

$$b_C^{(s)} = \langle K_C^*, R_s K_C \rangle.$$

At that birth stage we impose

$$(4.4) \quad |b_C^{(s)} - \lambda_C(R_s)| < \eta_C^{(s)},$$

and, for its two children C_+, C_- ,

$$(4.5) \quad |\lambda_{C_\pm}(R_s) - \lambda_C(R_s)| < \rho_{C, C_\pm}^{(s)}.$$

All other finite admissible side constraints whose newest inner block is born at this stage are included in the same application of the lemma.

We now bound the actual glued rooted-tree variation. The contribution of the already built old tree is the first two terms in G_s . If a birth root $B \in \mathcal{R}_s$ has no old parent, its root contribution satisfies

$$(4.6) \quad |b_B^{(s)}| \leq |\lambda_B(R_s)| + \eta_B^{(s)}.$$

If B has old parent $p(B)$, the gluing edge from the old tree into the future forest satisfies

$$(4.7) \quad |b_{p(B)}^{(s)} - b_B^{(s)}| \leq |b_{p(B)}^{(s)}| + |\lambda_B(R_s)| + \eta_B^{(s)},$$

and the old endpoint term $|b_{p(B)}^{(s)}|$ is included in the last sum defining G_s , with the correct multiplicity. Finally, for every future forest edge $C \rightarrow C'$,

$$(4.8) \quad |b_C^{(s)} - b_{C'}^{(s)}| \leq |b_C^{(s)} - \lambda_C(R_s)| + |\lambda_C(R_s) - \lambda_{C'}(R_s)| + |\lambda_{C'}(R_s) - b_{C'}^{(s)}| \leq \eta_C^{(s)} + \rho_{C, C'}^{(s)} + \eta_{C'}^{(s)}.$$

Summing (4.6)–(4.8), together with the old contribution contained in G_s , counts every old endpoint exactly with the multiplicity built into G_s , counts every birth-root trace within the harmless factor in (4.3), counts every future edge budget once, and counts every future node-error term no more than four times. Therefore the full glued variation of the strip is $< \tau_s$.

The scheduling is explicit and locally finite. For each strip s , enumerate the future nodes and future edges in $\mathcal{F}_s$ and choose the budgets in (4.3) before the construction starts. The node estimate (4.4) is imposed exactly at the stage when C is born, and the two child-trace estimates (4.5) are imposed at that same split, thereby registering the two outgoing edges from C . The locally finite admissible side constraints are enumerated by the half-step at which their newest block is born, assigned independent summable tolerances, and inserted into that finite splitting list. Since only finitely many blocks and strips exist at each finite stage, every application of Lemma 5.5 remains finite, and every node, edge, and admissible side constraint is imposed exactly once. The glued coefficient tree for each strip has no terminal leaves: every current frontier node is followed by its full future descendant tree. Lemma 5.7 therefore gives the stated L_1 multiplier norm bound, and summability of (τ_s) gives the total bound. $\square$

6. BIRTH OF STRIPS

Lemma 6.1 (Birth of outer multiplier strips). *Let A be an outer terminal atom with an infinite tail S_A , and let $\mathcal{U}_A$ be a free ultrafilter containing S_A . Let K, K' be inner Haar blocks and put*

$$Q_{K, K'} = (\text{Id} \otimes K^*)T(\text{Id} \otimes K') \in \mathcal{L}(L_p^0).$$

If

$$|\langle K^*, \Omega_A^{T\mathcal{U}_A} K' \rangle| < \sigma,$$

then

$$S'_A = S_A \cap \{m : |\alpha_m^A(Q_{K,K'})| < \sigma\}$$

belongs to $\mathcal{U}_A$ and is infinite. The same assertion holds for finitely many pairs by finite intersection inside $\mathcal{U}_A$.

Proof. For every m ,

$$\alpha_m^A(Q_{K,K'}) = \langle K^*, \Omega_{A,m}^T K' \rangle.$$

Taking the weak ultralimit along $\mathcal{U}_A$ gives the displayed scalar, so the set on which the inequality holds belongs to $\mathcal{U}_A$. $\square$

Lemma 6.2 (Birth of inner L_1 multiplier strips). *Let $B_1, \dots, B_d$ be the current inner terminal atoms. Let H, H' be outer Haar blocks and put*

$$R_{H,H'} = (H^* \otimes \text{Id})T(H' \otimes \cdot) \in \mathcal{L}(L_1^0).$$

If

$$\sum_{a=1}^d |\langle H^*, \mathcal{I}_{B_a}^T H' \rangle| < \tau,$$

then

$$\sum_{a=1}^d |\lambda_{B_a}(R_{H,H'})| < \tau.$$

Proof. This is the definition of $\mathcal{I}_B^T$ applied to the operator $(H^* \otimes \text{Id})T(H' \otimes \cdot)$. $\square$

7. THE STRIP-NORM LOCKED ARBITRARY-TO-MULTIPLIER REDUCTION

Theorem 7.1 (Locked arbitrary-to-multiplier reduction). *For every $T \in \mathcal{L}(X_{00})$ and every $\varepsilon > 0$, there are locked faithful outer and inner Haar systems $(H_I)_{I \in \mathcal{D}}$ and $(K_J)_{J \in \mathcal{D}}$, exterior maps $A, B \in \mathcal{L}(X_{00})$, and a bounded product Haar multiplier M on X_{00} such that*

$$AB = \text{Id}_{X_{00}}, \quad \|A\| \|B\| \leq 1,$$

and

$$\|ATB - M\| < \varepsilon.$$

The multiplier is given by the retained diagonal coefficients

$$Mu_{I,J} = d_{I,J}u_{I,J}, \quad d_{I,J} = \langle H_I^* \otimes K_J^*, T(H_I \otimes K_J) \rangle.$$

Proof. Fix the constants C_p , C , C_0 , and C_F appearing in the outer unconditional, inner strip, bookkeeping, and forest estimates below. Then choose three positive summable budget families, indexed by ordered pairs and ordered quadruples,

$$\sum_{J \neq J'} \sigma_{J,J'} < \frac{\varepsilon}{12C_p}, \quad \sum_{I \neq I'} \tau_{I,I'} < \frac{\varepsilon}{12C},$$

and

$$\sum_{I \neq I', J \neq J'} \mu_{I,I',J,J'} < \frac{\varepsilon}{12}.$$

All constants appearing in the one-parameter estimates below, including the Haar unconditional constant C_p , the forest multiplier constant C_F , and the bookkeeping constant C_0 , are now fixed once and for all. The following subdivisions are chosen after these constants are fixed and before the recursion begins. For every ordered inner pair $s = (J, J')$ put

$$\sigma_s^{\text{old}} = \sigma_s/4, \quad \sigma_s^{\text{tail}} = \sigma_s/4,$$

and choose positive numbers $\delta_{s,n}$, $n \geq 1$, with

$$(7.1) \quad \sum_{n=1}^{\infty} \delta_{s,n} < \sigma_s/4.$$

At the birth of the same-outer strip s , all already existing coefficients in that strip will be made $< \sigma_s^{\text{old}}$, while the shadow constraint will create a tail on which the trace is $< \sigma_s^{\text{tail}}$. The numbers $\delta_{s,n}$ are charged along the successive later outer splittings of atoms carrying s ; along any branch the accumulated deterioration is at most $\sum_n \delta_{s,n}$. Hence every future coefficient in the strip is $< \sigma_s^{\text{tail}} + \sum_n \delta_{s,n} < \sigma_s/2$, and the whole scalar strip has coefficient supremum $< \sigma_s$.

For every ordered outer pair $r = (I, I')$ put

$$(7.2) \quad \tau_r^{\text{old}} = \frac{\tau_r}{4C_0}, \quad \tau_r^{\text{root}} = \frac{\tau_r}{4C_0}.$$

At the birth of the same-inner strip r , the finite old charge G_r will be made $< \tau_r^{\text{old}}$ and the birth-root trace sum will be made $< \tau_r^{\text{root}}$. Thus the hypothesis

$$G_r + \sum_{B \in \mathcal{R}_r} |\lambda_B(R_r)| < \tau_r/C_0$$

of Proposition 5.8 holds with room to spare. The proposition then chooses the future node and edge budgets for that strip. Finally, each fully mixed ordered quadruple has the single tolerance

$$(7.3) \quad |\langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_{J'}) \rangle| < \mu_{I, I', J, J'}.$$

No other hidden budget is used; every side-constraint tolerance below is one of (7.1), (7.2), (7.3), or one of the node/edge tolerances chosen explicitly in Proposition 5.8.

We build the two faithful trees in generations, splitting every current terminal atom once before moving to the next generation. The half-steps are totally ordered:

$$\text{outer}(0) < \text{inner}(0) < \text{outer}(1) < \text{inner}(1) < \dots$$

Once an outer block H_I or an inner block K_J has been chosen, it is never changed. Each block receives its birth half-step $o(I)$ or $i(J)$. Before the recursion starts we also fix the following deterministic registration rule. Product-diagonal coefficients $(I, J) = (I', J')$ are never put into the error; they are retained in the multiplier. Every other ordered coefficient is assigned to exactly one registry time as follows:

- (R1) If $I = I'$ and $J \neq J'$, the ordered inner pair (J, J') is registered at time $\max(i(J), i(J'))$. The same-outer strip born then contains all coefficients with that ordered inner pair.
- (R2) If $I \neq I'$ and $J = J'$, the ordered outer pair (I, I') is registered at time $\max(o(I), o(I'))$. The same-inner strip born then contains all coefficients with that ordered outer pair.
- (R3) If $I \neq I'$ and $J \neq J'$, the single coefficient is registered when the last of the four blocks $H_I, H_{I'}, K_J, K_{J'}$ is born. Since the half-steps are totally ordered, this time is unique. If the last birth is an outer half-step the coefficient is placed in the outer side list; if it is an inner half-step it is placed in the inner side list. Simultaneous births inside one half-step are all treated as new blocks at that same registry time.

For (R1) and (R2), the registry time is the birth time of the whole strip, not the time at which every scalar coefficient in that strip is individually imposed. At strip birth, the coefficients whose active blocks already exist are imposed as finite admissible side constraints, and the coefficients involving later blocks are assigned to the corresponding one-parameter strip schedule. For (R3), the displayed scalar coefficient itself is imposed at its unique registry half-step.

At each finite stage only finitely many blocks, terminal atoms, active strips, and admissible side constraints are present. No side list ever asks for a self-diagonal coefficient, because those are either retained product-diagonal coefficients or are handled by the trace estimates in the

one-parameter lemmas. All countable side schedules used below are locally finite in the sense of Definition 4.2.

We maintain the following induction invariant at the beginning of every outer half-step.

- (I1) For each current outer terminal atom A there is a single infinite set $S_A \subset \mathbb{N}$ such that, for every registered same-outer strip active below A , the corresponding outer trace bound holds for all $m \in S_A$ with that strip's current allowance. The remaining future allowance increments for each such strip are bounded by the preassigned tail sum in (7.1).
- (I2) For each already registered same-inner strip, the finite old charge and the birth-root trace budget have been fixed as in (7.2), the future node and edge budgets from Proposition 5.8 have been assigned, and all node and edge requirements whose corresponding inner nodes have already been born have been imposed exactly once.
- (I3) Every fully mixed coefficient whose registry time is before the current half-step has been imposed exactly once with the tolerance (7.3).
- (I4) All admissible side constraints scheduled before the current half-step have been imposed, and no later side constraint involves only already born active blocks.

The initial invariant is empty. The two half-steps below preserve it: the outer half-step advances the common outer tails and registers the same-inner strips born from new outer pairs; the inner half-step advances the registered L_1 strips and registers the same-outer strips born from new inner pairs.

Outer half-step. Assume the current inner tree has been chosen. For every current outer terminal atom A , the invariant supplies one infinite common tail S_A for all already registered same-outer strips active below A . For such a strip $s = (J, J')$ write

$$Q_{J,J'} = (\text{Id} \otimes K_J^*)T(\text{Id} \otimes K_{J'}).$$

On the same set S_A , every active trace of $Q_{J,J'}$ is below its current allowance. Apply Lemma 5.1 simultaneously to this finite active family on the single common tail S_A , using the next unused increment $\delta_{s,n}$ from (7.1), and to the following finite side list.

- (a) For every ordered quadruple with $I \neq I'$ and $J \neq J'$ whose registry time (R3) is this outer half-step, impose the mixed inequality

$$(7.4) \quad |\langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_{J'}) \rangle| < \mu_{I,I',J,J'}.$$

Here J, J' are already born and at least one of I, I' is newly born.

- (b) For every ordered outer pair $r = (I, I')$ with $I \neq I'$ whose registry time (R2) is this outer half-step, let $\mathcal{T}_r^{\text{old}}$ be the finite inner Haar tree already built and let $\mathcal{R}_r$ be its current terminal frontier. For each old inner node $J \in \mathcal{T}_r^{\text{old}}$ put

$$b_J^{(r)} = \langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_J) \rangle.$$

Let

$$(7.5) \quad \omega_J^{(r)} = \mathbf{1}_{\{J=C_{\text{root}}\}} + \#\{\text{old edges of } \mathcal{T}_r^{\text{old}} \text{ incident with } J\} + \#\{B \in \mathcal{R}_r : p(B) = J\}.$$

Choose positive scalar tolerances $\eta_J^{(r)}$ for $J \in \mathcal{T}_r^{\text{old}}$ so that

$$\sum_{J \in \mathcal{T}_r^{\text{old}}} \omega_J^{(r)} \eta_J^{(r)} < \tau_r^{\text{old}},$$

and impose the finite family of scalar admissible inequalities $|b_J^{(r)}| < \eta_J^{(r)}$. Consequently

$$(7.6) \quad \sum_{J \in \mathcal{T}_r^{\text{old}}} \omega_J^{(r)} |b_J^{(r)}| < \tau_r^{\text{old}}.$$

Then the old charge G_r of Proposition 5.8 satisfies $G_r \leq \sum_J \omega_J^{(r)} |b_J^{(r)}| < \tau_r^{\text{old}}$, because every old edge term is bounded by the sum of the two endpoint moduli and the gluing endpoint multiplicities are exactly the last term in (7.5).

- (c) For every newly registered ordered outer pair $r = (I, I')$, choose positive scalar tolerances $\rho_B^{(r)}$, $B \in \mathcal{R}_r$, with $\sum_{B \in \mathcal{R}_r} \rho_B^{(r)} < \tau_r^{\text{root}}$, and impose for every current inner terminal atom $B \in \mathcal{R}_r$ the scalar admissible inequality

$$|\langle H_I^*, \mathcal{I}_B^T H_{I'} \rangle| < \rho_B^{(r)}.$$

Thus

$$(7.7) \quad \sum_{B \in \mathcal{R}_r} |\langle H_I^*, \mathcal{I}_B^T H_{I'} \rangle| < \tau_r^{\text{root}}.$$

We now verify admissibility, case by case. If the inner blocks $K_J, K_{J'}$ are frozen, the coefficient in (7.4) has the active outer form

$$\langle H_I^*, S_{J,J'} H_{I'} \rangle, \quad S_{J,J'} x = (\text{Id} \otimes K_J^*) T(x \otimes K_{J'}) \in \mathcal{L}(L_p^0).$$

The same formula with $J = J'$ gives each old same-inner coefficient $b_J^{(r)}$ in (7.6). The trace-root constraint (7.7) has the active outer form

$$\langle H_I^*, \mathcal{I}_B^T H_{I'} \rangle$$

with the bounded operator $\mathcal{I}_B^T \in \mathcal{L}(L_p^0)$. In all three cases $I \neq I'$ and at least one of the two outer blocks is newly born. If exactly one is new, the expression is one-sided; if both are new, the two new blocks are supported on distinct current outer atoms, so the expression is bilinear between distinct new blocks. Thus every constraint in (a)–(c) is admissible in the sense of Definition 4.1, and no outer self-diagonal side constraint occurs.

By Lemma 6.2, (7.7) says that for

$$R_r = (H_I^* \otimes \text{Id}) T(H_{I'} \otimes \cdot)$$

one has $\sum_{B \in \mathcal{R}_r} |\lambda_B(R_r)| < \tau_r^{\text{root}}$. Together with (7.6), this is the hypothesis (4.2) of Proposition 5.8; hence the same-inner strip r is registered and its future node and edge requirements are scheduled by that proposition. Lemma 5.1 passes the common tails of the already registered same-outer strips to the two children of every current outer terminal atom, with the accumulated increments charged to (7.1). Thus the induction invariant is preserved through the outer half-step.

Inner half-step. Now split all current inner terminal atoms. The active same-inner strips for already registered ordered outer off-diagonal pairs are advanced using Proposition 5.8; at the finite splitting level this is implemented by Lemma 5.5. The finite side list consists of the following constraints.

- (a) For every ordered quadruple with $I \neq I'$ and $J \neq J'$ whose registry time (R3) is this inner half-step, impose the mixed inequality

$$(7.8) \quad |\langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_{J'}) \rangle| < \mu_{I,I',J,J'}.$$

Here I, I' are already born and at least one of J, J' is newly born.

- (b) For every ordered inner pair $s = (J, J')$ with $J \neq J'$ whose registry time (R1) is this inner half-step, impose all already existing same-outer coefficients

$$(7.9) \quad |\langle H_I^* \otimes K_J^*, T(H_I \otimes K_{J'}) \rangle| < \sigma_s^{\text{old}}, \quad I \text{ already born.}$$

- (c) For every current outer terminal atom A , choose a free ultrafilter $\mathcal{U}_A$ containing the current common tail of the active outer strips below A , and impose

$$(7.10) \quad |\langle K_J^*, \Omega_A^{T, \mathcal{U}_A} K_{J'} \rangle| < \sigma_s^{\text{tail}}$$

for every newly registered ordered inner pair $s = (J, J')$.

Again the admissibility verification is explicit. For (7.8), with I, I' frozen, set

$$R_{I,I'}g = (H_I^* \otimes \text{Id})T(H_{I'} \otimes g) \in \mathcal{L}(L_1^0).$$

Then the coefficient is $\langle K_J^*, R_{I,I'}K_{J'} \rangle$. The same formula with $I = I'$ gives every old same-outer coefficient in (7.9). For (7.10), Proposition 3.2 gives a genuine bounded operator $\Omega_A^{T, \mathcal{U}_A} \in \mathcal{L}(L_1^0)$, and the constraint is $\langle K_J^*, \Omega_A^{T, \mathcal{U}_A}K_{J'} \rangle$. In all three cases $J \neq J'$ and at least one of $K_J, K_{J'}$ is newly born. If exactly one is new, the expression is one-sided; if both are new, the two new blocks are supported on distinct current inner atoms, so the expression is bilinear between distinct new blocks. Thus every side constraint in (a)–(c) is admissible, and no inner self-diagonal side constraint occurs. The self-diagonal expressions $\langle K_C^*, R_C K_C \rangle$ required for active same-inner strips are not side constraints; they are precisely the trace estimates supplied by Lemma 5.5.

Lemma 6.1, applied to the finite family of newly registered ordered inner pairs and then intersected inside the chosen ultrafilter, provides for every current outer terminal atom a new common infinite tail on which the newly registered traces are $< \sigma_s^{\text{tail}}$ and the old active traces remain within their updated allowances. The same-outer strips s are thereby registered, with their old coefficients controlled by (7.9) and their future coefficients controlled by Proposition 5.2 using the increments (7.1). The inner strip schedules have advanced by exactly one locally finite step, every mixed coefficient whose registry time is this half-step has been imposed with (7.8), and the induction invariant is preserved through the inner half-step.

Repeating the two half-steps gives locked faithful outer and inner Haar systems, and the invariant gives the claimed unique registration and scheduled strip estimates for every coefficient. Lemma 2.6 gives exterior maps A, B with $AB = \text{Id}_{X_{00}}$ and $\|A\| \|B\| \leq 1$. On the algebraic product-Haar span, let M_0 denote the diagonal multiplier with retained coefficients

$$M_0 u_{I,J} = \langle H_I^* \otimes K_J^*, T(H_I \otimes K_J) \rangle u_{I,J}.$$

At this point M_0 is only an algebraic diagonal operator; its bounded extension will follow after the off-diagonal part has been estimated.

By the deterministic assignment rule (R1)–(R3), product-diagonal coefficients $(I, J) = (I', J')$ are retained in M_0 . Each same-outer coefficient with $I = I'$ and $J \neq J'$ is assigned to exactly one same-outer strip, namely the strip born with the ordered inner pair (J, J') ; coefficients already present at that birth are imposed as finite admissible side constraints, and coefficients involving later outer blocks are controlled by the scheduled outer common-tail extension. Each same-inner coefficient with $I \neq I'$ and $J = J'$ is assigned analogously to exactly one same-inner strip born with the ordered outer pair (I, I') ; its old part is imposed at birth and its later inner coefficients are controlled by the scheduled L_1 strip extension. Finally, by (R3), each fully mixed coefficient is imposed individually at the unique half-step at which its last block is born. Thus the three strip classes are disjoint, every off-product-diagonal coefficient is assigned exactly once, and no coefficient is missed or charged twice.

We now build the off-diagonal operator directly. Its product Haar matrix is the matrix of ATB with the product diagonal removed. The following three operator-norm convergent sums define that off-diagonal operator.

First, for each ordered inner off-diagonal pair $J \neq J'$, collect the same-outer strip

$$R_{J,J'}^{\text{out}} = \sum_I \langle H_I^* \otimes K_J^*, T(H_I \otimes K_{J'}) \rangle (e_I \otimes e_I^*) \otimes (f_J \otimes f_{J'}^*).$$

The finitely many outer coefficients already present when (J, J') was born were controlled by inner half-step item (b), and all later outer coefficients are controlled by Proposition 5.2. Thus the scalar outer Haar multiplier has norm at most $C_p \sigma_{J,J'}$. The strip acts by first applying the

inner coefficient map $g \mapsto \langle f_{J'}^*, g \rangle$, then the scalar outer multiplier, and finally the embedding $a \mapsto af_J$; both inner rank-one maps have norm one. Hence

$$\sum_{J \neq J'} \|R_{J,J'}^{\text{out}}\| < \varepsilon/4.$$

Second, for each ordered outer off-diagonal pair $I \neq I'$, collect the same-inner strip

$$R_{I,I'}^{\text{in}} = (e_I \otimes e_{I'}^*) \otimes \sum_J \langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_J) \rangle (f_J \otimes f_J^*).$$

The outer slice $F \mapsto \int e_{I'}^*(s)F(s, \cdot) ds$ has norm one from $L_p(L_1)$ to L_1 , and the embedding $g \mapsto e_I \otimes g$ also has norm one. For this ordered pair, outer half-step item (b) supplied the finite old charge $G_{I,I'}$, item (c) supplied the birth-root lambda condition, and Proposition 5.8 controls the whole later forest including gluing edges. By Lemma 5.7, equivalently the Semenov–Uksusov estimate (4.1) applied after the zero-extension reduction, the inner multiplier norm is at most $C\tau_{I,I'}$. Hence

$$\sum_{I \neq I'} \|R_{I,I'}^{\text{in}}\| < \varepsilon/4.$$

Third, the mixed part is

$$R^{\text{mix}} = \sum_{I \neq I', J \neq J'} \langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_{J'}) \rangle (e_I \otimes e_{I'}^*) \otimes (f_J \otimes f_{J'}^*).$$

Every product rank-one has norm one. By (R3), each fully mixed ordered coefficient has a unique registry half-step; if that half-step is outer the inequality (7.4) is imposed, and if it is inner the inequality (7.8) is imposed. Hence for every $I \neq I'$ and $J \neq J'$,

$$|\langle H_I^* \otimes K_J^*, T(H_{I'} \otimes K_{J'}) \rangle| < \mu_{I,I',J,J'}.$$

Therefore the mixed series converges absolutely in operator norm and

$$\|R^{\text{mix}}\| < \varepsilon/4.$$

Let

$$O = \sum_{J \neq J'} R_{J,J'}^{\text{out}} + \sum_{I \neq I'} R_{I,I'}^{\text{in}} + R^{\text{mix}}.$$

The preceding estimates and the initial choices of the three summable budget families give $\|O\| < \varepsilon$. On the dense product-Haar span, the matrix coefficients of O are exactly the off-product-diagonal coefficients of ATB , while the algebraic operator M_0 has exactly the product-diagonal coefficients. Coefficient functionals separate points in X_{00} , hence

$$ATB = M_0 + O$$

on the product-Haar span. Therefore $M := ATB - O$ is a bounded operator on X_{00} extending M_0 , and M is the required bounded product Haar multiplier with

$$\|ATB - M\| = \|O\| < \varepsilon.$$

□

8. SCALAR COMPRESSION AND PRIMARITY

Theorem 8.1 (Scalar compression on X_{00}). *For every $S \in \mathcal{L}(X_{00})$ and every $\varepsilon > 0$, there are $A, B \in \mathcal{L}(X_{00})$ and a scalar c such that*

$$AB = \text{Id}_{X_{00}}, \quad \|ASB - c\text{Id}_{X_{00}}\| < \varepsilon.$$

Proof. Apply Theorem 7.1 to S with accuracy $\delta > 0$. We obtain A_0, B_0 and a bounded product Haar multiplier M such that

$$A_0B_0 = \text{Id}_{X_{00}}, \quad \|A_0SB_0 - M\| < \delta.$$

Apply Theorem 2.4 to M with accuracy δ . There are A_1, B_1 and a scalar c such that

$$A_1B_1 = \text{Id}_{X_{00}}, \quad \|A_1MB_1 - c\text{Id}_{X_{00}}\| < \delta, \quad \|A_1\| \|B_1\| \leq 1 + \delta.$$

Set

$$A = A_1A_0, \quad B = B_0B_1.$$

Then $AB = A_1A_0B_0B_1 = A_1B_1 = \text{Id}_{X_{00}}$, and

$$\|ASB - c\text{Id}\| \leq \|A_1\| \|B_1\| \delta + \delta.$$

Choosing $\delta > 0$ so that $(1 + \delta)\delta + \delta < \varepsilon$ gives the result. $\square$

Corollary 8.2 (Primarity of $L_p(L_1)$). *For every $1 < p < \infty$, the space $L_p([0, 1]; L_1[0, 1])$ is primary.*

Proof. This is Proposition 1.2 applied to Theorem 8.1. $\square$

ARCHITECTURE OF THE PROOF

The proof uses the locked strip-norm route with locally finite admissible side schedules. The local L_1 trace theorem is obtained from an explicit Reynolds average and the terminal Semenov–Uksusov branch-variation term. The outer child-tail preservation keeps the diagonal and finite admissible side-constraint event inside the expected-count/pigeonhole argument. The L_1 strip estimate uses branch variation with explicit old-tree, gluing, root, node-error, and edge-error budgets. Finally, the birth-time rule in Theorem 7.1 assigns every off-product-diagonal coefficient exactly once, either to a strip schedule or to an individual mixed side constraint. Thus diagonal product coefficients are retained as a multiplier, same-outer strips are controlled by L_p unconditionality, same-inner strips are controlled by L_1 branch variation, and genuinely mixed coefficients are made absolutely summable. The external inputs are isolated in Theorem 2.1, Theorem 2.4, and the faithful exterior-map Lemma 2.6.

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